

1 Explainability in Reinforcement Learning

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Chapter 1

Explainability in Reinforcement Learning

Today, the majority of Artificial Intelligence (AI) agents with which humans interact are deployed in controlled or low-risk environments. For example, agents capable of playing games such as Go operate in settings where potential errors remain limited in terms of consequences [Silver et al., 2017]. Even more autonomous systems, such as agents based on Large Language Model (LLM), remain constrained by human supervision mechanisms and are often restricted to tasks such as code generation or information retrieval [Chen et al., 2021, Lewis et al., 2021]. Safety-critical applications integrating machine learning components are still rare. Introducing autonomous agents in such environments, requires the establishment of strong guarantees [Seshia et al., 2022]. These requirements include in particular:

- validating the system’s functioning before deployment,
- identifying the origin of a malfunction in case of failure,
- and certifying that an observed failure will not occur again.

The main challenge lies in the fact that the performing agents currently available rely on both Reinforcement Learning (RL) and Deep Learning (DL) [Mnih et al., 2013]. As a result, the mechanisms underlying their decisions cannot be understood directly from the learned model. This lack of explainability makes their use difficult in safety-critical contexts. To address this limitation, the field of eXplainable Reinforcement Learning (XRL) has emerged as a subfield of eXplainable Artificial Intelligence (XAI). The objective of XRL is to develop explainability methods that make it possible to obtain explanations of RL agents. Due to the relevance of the topic and to foster broader real-world adoption of RL, a growing number of research teams are working on developing new explainability methods. This enthusiasm is reflected in a rapid increase in the number of publications, as illustrated in Figure 1.1.¹

¹The complete search protocol, including the databases consulted and the full set of query terms, has been archived on Zenodo <https://zenodo.org/records/19387785>.

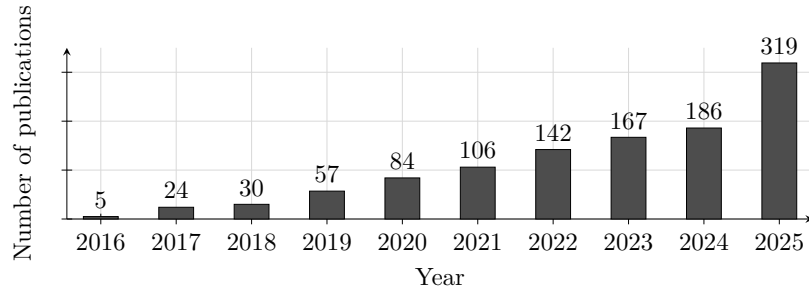


Figure 1.1: Yearly growth of Explainable Reinforcement Learning research with at least one citation on arXiv.

The growing number and diversity of these methods make it increasingly difficult to obtain a unified view of the field. This motivated our work on developing a unified taxonomy of explainability methods for RL, initially presented in [Alaarabiou et al., 2024]. The taxonomy provides a common structure for organizing existing approaches according to the type of information they use and the form of explanation they provide. Building on this work, this chapter provides a detailed presentation of the proposed taxonomy and uses it to organize a broader state-of-the-art review of XRL.

This chapter is organized as follows:

- Section 1.1 begins by clarifying what we mean by an explanation in the context of XAI.
- Section 1.1 then introduces a taxonomy for categorizing existing XRL methods.
- Section 1.3 provides a comprehensive state-of-the-art review structured around this taxonomy.
- Section 1.4 concludes the chapter with a summary and discussion.

1.1 Explanation in Artificial Intelligence

In this section, we define what an explanation is in the context of XAI and how it can be obtained. We first introduce the concept of explanation in general. We then describe how this concept applies to XAI. Finally, we further detail the properties that characterize an explainability method in the context of XAI.

1.1.1 Explanation

Defining what constitutes an explanation is challenging because the concept is inherently multidisciplinary. When we refer to explanation, we are inherently referring to an explanation provided to a human. As soon as humans are involved, the problem is no longer purely computational or mathematical. It involves several fields of research, such as philosophy, psychology, and sociology. So far, these disciplines have not converged on a unified definition of explanation that can be directly applied to XAI [Miller, 2019, Bellucci et al., 2021].

For this reason, we propose the following definition of explanation. An explanation involves two entities: the object (Definition 1.1.1) being explained and the receiver of the explanation. The receiver is a human who aims to understand how the object operates and will be referred to as the observer throughout this manuscript (Definition 1.1.2). The observer seeks to understand the object and, in doing so, builds a mental model of its functioning. An explanation is therefore what allows the observer to refine and correct their mental model of how the object operates (Definition 1.1.4). This refinement is obtained through a transformation process that produces an explanation from the object being explained. We refer to this process as an explainability method (Definition 1.1.5), detailed further in Section 1.1.3. For an observer to understand an explanation, the information it contains must be interpretable (Definition 1.1.3). Information is considered interpretable when it can be directly understood by a human without requiring any additional transformation process.

Definition 1.1.1: Object

The object of an explanation is the system the observer wants to understand.

Definition 1.1.2: Observer

The observer is the human who aims to understand an object. The observer possesses a mental model of the object being explained. This mental model is an internal representation of how the observer believes the system operates. This representation may be incomplete or inaccurate.

Definition 1.1.3: Interpretable

Information is interpretable when it can be directly understood by an observer without requiring any additional transformation process.

Definition 1.1.4: Explanation

An explanation is any interpretable information that enables an observer to refine or correct their mental model of the object of the explanation.

Definition 1.1.5: Explainability Method

An explainability method is a transformation process that produces an explanation from the object being explained.

This definition of explanation provides the foundation for the approach to explainability adopted throughout this manuscript. Explainability is not regarded as a binary property but as a continuum that evolves with the development of scientific knowledge [Siegel and Craver, 2024, Zednik, 2019].

Accordingly, the opacity of a system should be interpreted as an epistemic limitation reflecting the current state of knowledge rather than as an intrinsic property of the system itself. This view is consistent with the methodological commitments of scientific inquiry. Scientific progress is driven by the continuous refinement of theories and analytical methods that provide progressively deeper accounts of observed phenomena. Consequently, explainability should

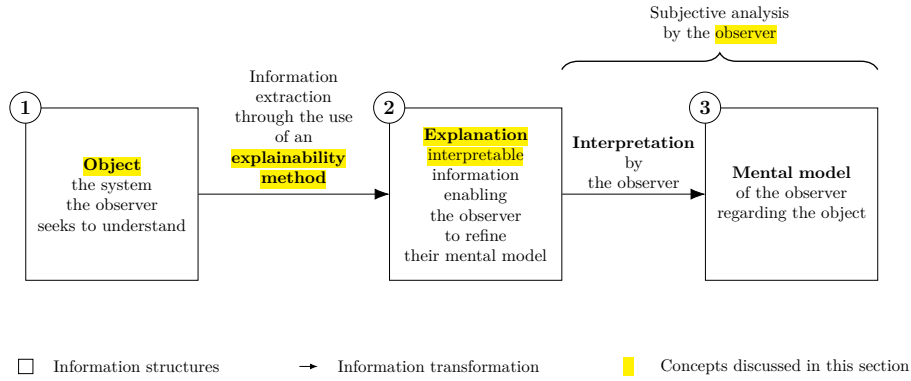


Figure 1.2: Explanation process.

be regarded as a matter of degree, increasing as the understanding of a system improves.

This quantitative perspective allows us to express an intuitive idea: not all explanations provide the same level of value [Doshi-Velez and Kim, 2017]. The effectiveness of an explanation depends on its ability to improve the observer’s understanding of the system. For example, describing a marginal aspect of a system is different from correcting an observer’s misunderstanding of a key mechanism.

Figure 1.2 summarizes this whole explanation process, from the object of the explanation to the mental model refined by the observer, as a chain of three information structures, depicted as boxes and connected by arrows denoting a transformation of information:

- (1) The object of the explanation, that is, the system the observer seeks to understand.
- (2) The explanation, the interpretable information produced from the object. This transformation is performed through the use of an explainability method.
- (3) The mental model held by the observer regarding the object.

A brace spanning boxes (2) and (3) indicates that this second half of the process, from the explanation to the resulting mental model, constitutes a subjective analysis carried out by the observer.

1.1.2 Explainable Artificial Intelligence

Now that the general framework of explanation has been established, we can examine how this concept applies in the context of AI. In the context of XAI, the object of explanation is an AI system. More precisely, XAI emerged as a response to the success of DL models [Gunning and Aha, 2019, Barredo Arrieta et al., 2020]. In these architectures, information is represented in a distributed manner across the network through patterns of activation and connections between computational units. Unlike symbolic approaches, individual components of Neural Network (NN) cannot be directly associated with human concepts. From this starting point, we identify two types of approaches:

- **Substitutive explainability:** In this approach, the underlying assumption is that there is nothing to explain within a NN. There is no semantic meaning associated with each computational component, and therefore nothing to interpret. This corresponds to the “black box” view of NN. Under this assumption, XAI consists of replacing as much of the DL system as possible with alternative AI methods [Rudin, 2019].
- **Intrinsic explainability:** In this approach, it is argued that NN inherently contain mechanisms that can be explained. The limitation is considered to lie not in the networks themselves, but in the absence of adequate tools to analyze their functioning [Olah et al., 2020].

This manuscript considers methods from both approaches in order to provide a comprehensive overview of the XAI field. The substitutive view has produced valuable results, particularly when interpretability is prioritized over the raw predictive performance of DL models [Rudin, 2019]. However, this manuscript adopts *Mechanistic Hypothesis* (Hyp. 1) [Siegel and Craver, 2024, Zednik, 2019], stated below.

Hypothesis 1: Mechanistic Hypothesis

Every AI system results from a sequence of causal computations. Because the rules governing these interactions are mechanical, the behavior of the system can be explained. If an AI system cannot be explained, this reflects a limitation of knowledge rather than an intrinsic property of the system.

Under this hypothesis, the opacity of an AI system reflects a limitation of current knowledge and analytical tools rather than an intrinsic property of the system, so the “black box” framing of NN is not adopted as an ontological claim. As a consequence, the perspective adopted throughout this manuscript is that of intrinsic explainability.

1.1.3 Explainability Method in XAI

Now that the general framework of XAI has been established, we need to further detail the properties of an explanation in this context. In our framework, an explanation in the field of XAI can be characterized by three components:

- **Source:** refers to the origin of the information used by the explainability method. The explainability method transforms the raw information produced by the AI system into an explanation that can be understood by an observer.
- **Form:** refers to the representation through which the explanation is communicated. The form defines how the information is presented to the observer and must allow human interpretation. An explanation can take many different forms, including text, images, computer code, and mathematical equations.
- **Subject:** refers to the aspect of the AI system that is being explained. In XAI, this distinction is commonly associated with the difference between local and global explanations. A local explanation describes why an AI system produced a specific output in a given situation. A global explanation aims to describe the general functioning of the AI system.

These three properties of an explanation are entirely determined by the explainability method (Definition 1.1.5) used to produce it. In this context, an explainability method is a transformation process that maps a source of information into an explanation characterized by a specific form and subject. Research in XAI focuses on developing new methods capable of performing this transformation in order to overcome the lack of interpretability associated with DL models.

1.2 Taxonomy of XRL Methods

Now that the concept of XAI has been defined, we turn to its application in RL, known as XRL. As in XAI, explanations in XRL are produced by explainability methods and can be described in terms of their subject, source, and form. This section reviews the possible variations of these three aspects in the specific context of RL.

1.2.1 Subject

The first question to address is the subject of explanation in the context of RL. To answer this question, it is necessary to identify the different components involved in a RL system and determine which of them constitutes the object of explanation. A RL system can be described through two main components: the environment and the policy.

The environment defines the dynamics in which the agent evolves, including the possible states, observations, actions, and transitions. Environments are simulations designed by humans, and their dynamics are explicitly defined. Therefore, it is assumed that the environment is known to the observer and does not require explanation.

186 The main object of interest is therefore the agent’s policy. The policy
 187 represents the decision-making mechanism of the agent. It is a function that maps
 188 observations to actions and is learned through interaction with the environment in
 189 order to maximize the expected cumulative reward. The policy is the component
 190 that encodes the learning process and that XRL aims to explain.

191 Explaining a policy introduces additional challenges compared with classical
 192 XAI settings. In traditional XAI problems, the objective is to explain an isolated
 193 decision, such as determining which features of an image led to a classification
 194 result. In contrast, RL introduces a temporal dimension into the decision-making
 195 process. An action is not only determined by the current observation but also
 196 influences future states and rewards. Therefore, understanding the behavior of
 197 an agent may require considering a sequence of interactions between the agent
 198 and its environment.

199 This temporal dimension leads to a distinction between two objectives of
 200 explanation in XRL:

- 201 • **Decision explanations:** focus on identifying the elements of the current
 202 observation that influenced the action selected by the policy. This type of
 203 explanation is close to classical XAI approaches, as it analyzes the rela-
 204 tionship between inputs and outputs while abstracting away the sequential
 205 nature of the decision process.
- 206 • **Strategy explanations:** aim to characterize the agent’s behavior over
 207 time. They focus on sequences of decisions and interactions with the
 208 environment in order to understand how the agent achieves its long-term
 209 objective. This type of explanation is specific to RL, as it addresses the
 210 temporal and goal-oriented nature of the learning process.

211 These two objectives should not be considered as strictly separated categories,
 212 but rather as two extremes of a continuum. Understanding the elements of an
 213 observation that influence a specific decision can help the observer generalize
 214 this information to infer broader aspects of the agent’s behavior. Conversely,
 215 understanding the agent’s overall strategy can provide insights into the reasoning
 216 behind individual decisions.

217 1.2.2 Source

218 We identify three sources of explanation in RL, all obtained during or after the
219 learning process and containing information about the agent:

- 220 • **Policies:** policies are functions that generate a distribution over the action
221 space for a given observation. The objective of RL is to learn a policy that
222 maximizes the expected cumulative future rewards. During the learning
223 phase, the policy is iteratively improved through interactions with the
224 environment.
- 225 • **Trajectories:** trajectories represent sequences of successive interactions
226 between the agent and the environment. At each step, the agent selects
227 an action based on its observation, which modifies the environment and
228 produces a reward as well as a new observation. This process continues
229 until the end of the trajectory.
- 230 • **Value functions:** value functions provide predictions about the future
231 outcome of a trajectory from a given observation. The most commonly
232 used value function is the critic function, which estimates the expected
233 sum of future rewards. This prediction guides the learning process, as the
234 agent updates its policy to maximize the estimated future return. However,
235 other types of value functions can also be used as sources of explainability.

236 1.2.3 Form

237 We identify six forms of explanation in the RL setting:

- 238 • **Policy Model:** an interpretable model of the agent’s policy function.
- 239 • **Saliency Map:** a weighting of the input features representing their
240 influence on the output of the policy function.
- 241 • **Counterfactual:** a modified observation obtained by perturbing the
242 original observation in order to produce a different action from the agent.
- 243 • **Agent Trajectory:** a sequence of successive interactions between the
244 agent and the environment.
- 245 • **Trajectory Prediction:** a prediction about the future evolution of the
246 trajectory produced by a value function.
- 247 • **Interaction Model:** an interpretable model of the interactions between
248 the agent and the environment.

249 Among the three components characterizing an explanation, the form is the
250 one that varies most significantly across explainability methods. While subjects
251 and sources tend to be relatively stable in the XRL literature, the diversity in
252 how explanations are presented is considerably greater. The list above is by no
253 means definitive and may evolve as new approaches emerge. For this reason, the
254 following section is organized according to the form of explanations and presents
255 existing explainability methods based on this criterion.

Explanation Subject	Explanation Form	Explainability Method Family	Source of Explanation		
			Policy	Trajectories	Value functions
Decision	Policy Model	<i>Interpretable policy learning</i>	✓		
		<i>Policy approximation via interpretable model</i>	✓		
	Saliency Map	<i>Observation perturbation</i>	✓		
		<i>Observation gradient analysis</i>	✓		
		<i>Attention-based architecture</i>	✓		
	Counterfactual	<i>Latent space usage for counterfactual generation</i>	✓		
Strategy	Agent Trajectory	<i>Hierarchical decomposition</i>	✓	✓	
		<i>Extraction via critic analysis</i>		✓	✓
	Trajectory Prediction	<i>Critic enrichment</i>			✓
	Interaction Model	<i>Bayesian network learning</i>		✓	
		<i>Latent space usage for MDP generation</i>	✓		

Table 1.1: Taxonomy of explainability methods in reinforcement learning.

1.3 State-of-the-Art Review of XRL Methods

The purpose of this section is to detail the functioning of existing explainability methods (see Table 1.1). They are presented according to the form of explanation they produce, following the taxonomy introduced in Section 1.2.

A family of methods is defined as a set of methods that share the same source, form, and subject of explanation, as well as a similar procedure for generating explanations. Methods belonging to the same family may differ in their implementation or in the details of the generation process, while relying on the same underlying principle.

The remainder of this review is organized by form of explanation, following the six forms identified in Section 1.2.3. For each form, we first describe its underlying principle and its limitations, then present the families of methods that produce it.

1.3.1 Policy Model

The policy function can be represented by a wide variety of function classes. In modern RL, NN is the most commonly used models for representing policies. Although these models achieve the highest performance, their connectionist nature makes their internal representations non-interpretable.

For this reason, a substitutive approach to XRL aims to replace these models with alternative models considered interpretable. Such models can be understood through direct analysis of their representations. The main classes of interpretable models used to represent policy functions include decision trees [Topin et al., 2021], algorithms [Trivedi et al., 2022], formal logic [Payani and Fekri, 2019], and linear models [Garreau and von Luxburg, 2020].

Explanations derived from an interpretable model primarily provide insights into the decision-making process, since what is modeled is the mapping between

an observation and an action. If the model is sufficiently simple or well structured, the observer can form a mental representation of a sequence of decisions across the environment and thus obtain information about the agent’s strategy.

The main limitation of these methods is that interpretable models rely on symbolic representations. When the policy to be represented is too complex, achieving comparable performance requires a large number of rules or conditions. The resulting model can quickly become difficult to analyze and lose its interpretability.

For example, representing a policy capable of playing Go using a symbolic model would require considering a very large number of possible situations and specific cases. Even if such a representation were possible, the resulting model would likely become intractable for human analysis due to its complexity. This illustrates why NN-based approaches have become dominant in such domains, as they can learn complex decision strategies without requiring an explicit enumeration of rules.

To obtain an interpretable policy model, two families of explainability methods are available: *interpretable policy learning* and *policy approximation via interpretable model*.

Interpretable Policy Learning

This family of methods aims at learning a policy represented by an interpretable model [Topin et al., 2021, Trivedi et al., 2022]. It is the most direct approach to addressing the challenges posed by XRL. This type of training requires modifying the Markov Decision Processes (MDP) in order to adapt it to the search space and to the type of policy representation. Moreover, this search space often needs to be reduced to make learning feasible. Such a reduction requires the integration of expert knowledge into the learning process. With this family of methods, the *interpretable policy function* itself is used to provide explanations to the observer.

Policy Approximation via Interpretable Model

An interpretable model can be trained to approximate a policy learned through deep RL [Sieusahai and Guzdial, 2021]. In this setting, the problem is formulated as a supervised learning task, where the original deep RL policy is used to label each observation with the action it selects, such as the observation from the Atari game “Pong” shown in Figure 1.3a. Using this dataset, the interpretable model is trained to reproduce the behavior of the original policy, yielding for instance the decision tree illustrated in Figure 1.3b, obtained by approximating a deep RL policy trained on that environment. The resulting interpretable model is then used as an explanation for the observer. If this approximation is sufficiently faithful, it can also be deployed in the environment in place of the original policy, thereby providing a high degree of control over the agent.

1.3.2 Saliency Map

The influence of each part of the input on the action selected by the policy can be visualized using heatmaps (Figure 1.4). In the RL setting, the input corresponds to the observation received by the agent at a given time step. Heatmaps therefore

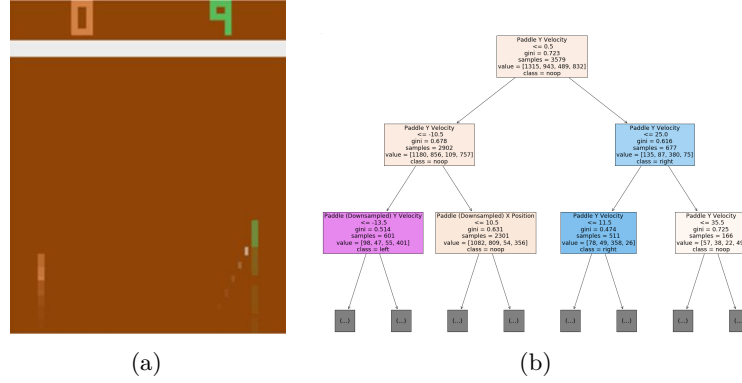


Figure 1.3: Policy approximation via an interpretable model [Siesahai and Guzdial, 2021].

highlight the regions of the observation that have the greatest influence on the action selection process. By providing this information, they allow the observer to identify which elements of the environment are considered important by the policy when making a decision. In Figure 1.4, both panels correspond to the same observation of the Atari game “Breakout”, with panel (a) obtained by observation perturbation and panel (b) by observation gradient analysis.

A limitation of heatmap-based methods is that the resulting explanations are not always directly meaningful to the observer. As illustrated in Figure 1.4b, interpreting which regions are important often requires human reasoning and involves a degree of subjectivity. This limitation becomes particularly relevant in RL, where the objective is not only to explain a single decision but also to understand the factors influencing a sequence of decisions over time.

Three families of explainability methods can be used to obtain this type of explanation: *observation perturbation*, *observation gradient analysis*, and *attention-based architectures*.

Observation Perturbation

Perturbation analysis consists in applying various modifications to the observation in order to study their impact on the output of the policy function [Greydanus et al., 2018]. The greater the variation in the output caused by a perturbation applied to a given component of the input vector, the higher the weight assigned to that component. The magnitude of the perturbation is measured by computing the distance between the original output and the output obtained from the perturbed input. Using this set of weights, a *heatmap* is constructed, enabling the observer to visualize which parts of the observation vector are most sensitive in the action-selection process (see Figure 1.4a).

Observation Gradient Analysis

Observation gradient analysis refers to a set of methods that require a differentiable policy function in order to extract information from it [Simonyan et al., 2014, Weiskamp et al., 2019, Huber et al., 2019]. To this end, the gradient of the policy function is computed with respect to all components of the input vector.

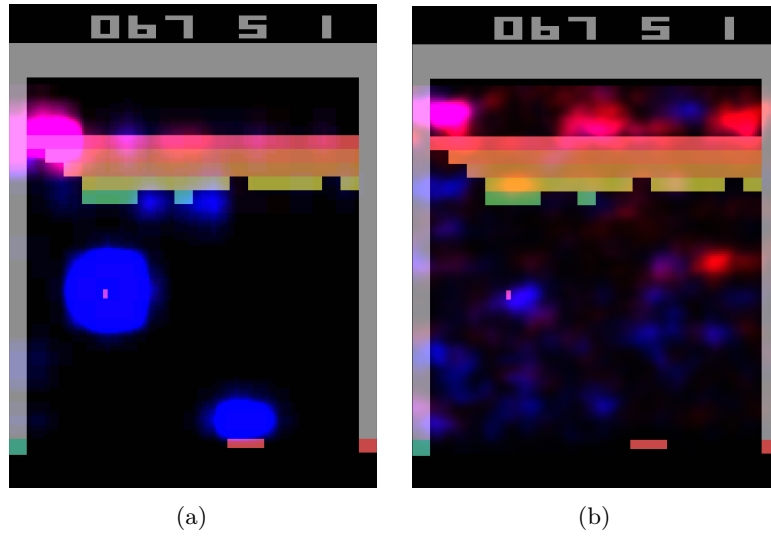


Figure 1.4: Heatmaps for the same observation of the Atari game “Breakout” [Greydanus et al., 2018].

For each input component, the resulting gradient provides a weighting that represents its importance in the output of the policy function. Based on these weightings, a *heatmap* is constructed to allow the observer to visually identify the parts of the input vector that are the most influential in the policy function’s output (see Figure 1.4b).

Attention-Based Architecture

Attention blocks constitute a NN architecture that makes it possible to weight the relative importance of one part of a vector with respect to the other parts of the same vector [Vaswani et al., 2023]. These weights are computed as a function of the vector itself and are learned by the NN during training. For a given observation, each component of the observation vector is assigned a scalar value. It can therefore be assumed that the components associated with the highest weights are the most influential in determining the output [Mott et al., 2019, Itaya et al., 2021]. This attention-based *heatmap* is then provided to the observer as the explanation, as illustrated for the Atari game “Breakout” in Figure 1.5a, where the highlighted regions are easily interpretable. However, as shown for “Ms. Pac-Man” in Figure 1.5b, the attention weights are not always concentrated on a small, easily interpretable set of regions, which can make this type of explanation difficult to exploit.

1.3.3 Counterfactual

A counterfactual explanation is a new observation, close to the original one, that leads the agent to select a different action. Comparing the two observations reveals which elements of the environment must change for the agent to behave differently, and thus provides insight into its decision-making process. Unlike the other forms of explanation, this form is currently associated with a single

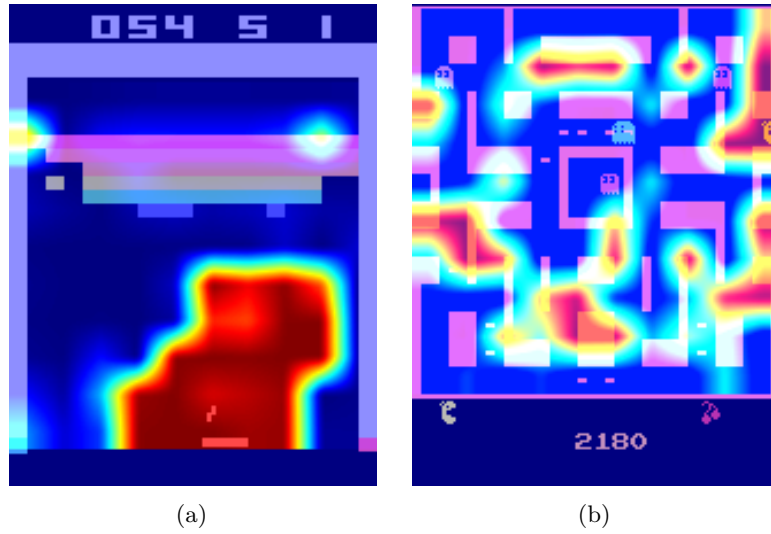


Figure 1.5: Attention-based heatmaps for the Atari games “Breakout” and “Ms. Pac-Man” [Mott et al., 2019].

381 family of methods, based on the *use of the Latent Space (LS) for counterfactual*
 382 *generation*.

383 *Latent Space Usage for Counterfactual Generation*

384 This family of methods uses generative models to build counterfactuals from
 385 the Latent Representation (LR) of the observation [Bond-Taylor et al., 2021,
 386 Olson et al., 2021]. The observation is encoded into the LS, its representation
 387 is modified until the agent selects the desired action, and this modified repre-
 388 sentation is decoded back into a new observation. Working in the LS rather
 389 than at the pixel level keeps the counterfactual close to the original state and
 390 consistent with the environment. The fidelity of the generated observations to
 391 the real game can then be assessed through user studies [Olson et al., 2021].

392 Figure 1.6 illustrates this approach on the Atari game *Space Invaders*
 393 [Olson et al., 2021]. In the original observation (Figure 1.6a) the agent se-
 394 lects “LEFT”, whereas in the counterfactual observation (Figure 1.6b) it selects
 395 “RIGHT_FIRE”. Here, removing an enemy is enough to change the action,
 396 which suggests that the position of that enemy influences the agent’s decision.
 397 Such observations remain only indicative. Identifying a reliable pattern requires
 398 repeating the analysis across many situations.

399 1.3.4 Agent Trajectory

400 Until now, the presented methods have focused on explaining individual decisions
 401 made by the agent. The methods considered from this point on aim to provide
 402 insight into the agent’s strategy by analyzing its behavior over time.

403 The most direct approach consists in visualizing agent trajectories, most
 404 commonly through complete episodes. By analyzing these trajectories, the
 405 observer can infer the dynamics underlying the agent’s behavior and how its

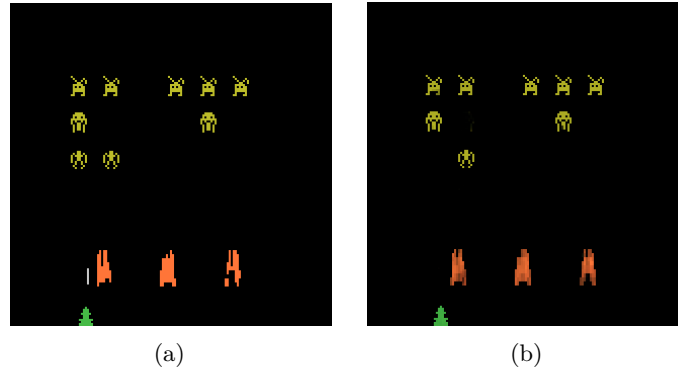


Figure 1.6: Counterfactual explanation of the agent’s behavior for the Atari game *Space Invaders* [Olson et al., 2021]. Panel a shows the original observation and panel b the counterfactual observation produced by a generative model.

decisions evolve through time. This type of visualization is not specific to XRL. In practice, visualizing agent episodes is a standard step in the development and evaluation of RL. It provides a simple way to verify that the agent behaves as expected and to detect potential implementation errors.

This simple approach can be further enhanced using families of methods such as *hierarchical decomposition* and *extraction via critic analysis*.

Extraction via Critic Analysis

In this family of methods, the predictions of the critic are used to determine which trajectories to present to the observer [Sequeira and Gervasio, 2020]. A set of states is selected from multiple trajectories. For a given state, all possible actions are examined. The state is assigned a value equal to the difference between the critic’s prediction for the lowest-valued action and the prediction for the highest-valued action. Trajectories containing the states with the highest values according to this procedure are then chosen to be presented to the observer. These trajectories are considered critical moments of the agent and are therefore relevant for analysis. For this family of methods, the *selected trajectories* represent the interpretable information provided to the observer.

Hierarchical Decomposition

Hierarchical agents are composed of a set of sub-policies specialized for specific tasks within subsets of the environment’s states. This family of methods distributes the optimization of the reward function across multiple models, significantly simplifying the learning process. Moreover, this decomposition of the policy function can also provide explainability in the agent’s strategy [Beyret et al., 2019]. If a sub-policy is selected, it indicates that the agent will execute a specialized behavior. This behavioral specialization provides insight into the strategy the agent employs to maximize the reward function. The agent’s *trajectory*, together with the *sub-policy* selected for each action, is used as interpretable information.

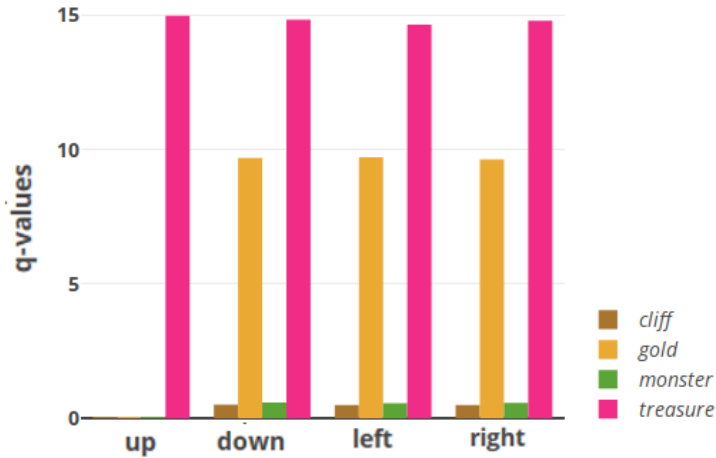


Figure 1.7: Prediction of sub-rewards as a function of the action selected by the agent for a given state [Juozapaitis et al., 2019].

1.3.5 Trajectory Prediction

Making predictions about the future of a trajectory allows one to form a representation of what the system expects, based on its training experience. These predictions are produced using value functions. Such functions provide insights into the future evolution of the agent’s trajectory. By their nature, this form of explanation informs the observer about the strategy adopted by the agent. Value functions provide an estimate of the expected cumulative reward from a given state. Consequently, a value function maps an observation to a scalar value in $\mathbb{R}$.

This aggregated prediction provides limited information about the underlying factors that contribute to this value. It does not directly reveal which aspects of the task are being optimized or how the predicted return is composed. Unlike the other forms of explanation, this form is currently associated with a single family of methods, aimed at improving the interpretability of value-based explanations: *critic enrichment*.

Critic Enrichment

In order to obtain interpretable information from the predictions of the critic, it is possible to decompose the critic into multiple sub-functions [Juozapaitis et al., 2019]. The reward function is thus decomposed into multiple sub-reward functions, each representing a sub-objective of the task to be accomplished. Similarly, for each of these sub-reward functions, corresponding sub-critic functions are defined, along with a function that combines their outputs. During training, the agent selects its actions so as to maximize this combination function of the sub-critics. Using this approach, for each observation and action, a prediction is obtained regarding the sub-objectives the agent is attempting to maximize, so that the action chosen by the agent reveals which sub-reward it aims to maximize (see Figure 1.7). This *prediction* is then used as the explanation for the observer.

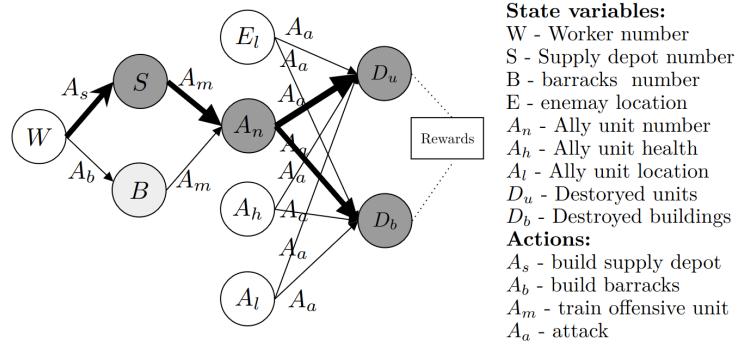


Figure 1.8: Agent-environment interaction model for the video game *StarCraft II* [Madumal et al., 2020].

1.3.6 Interaction Model

Modeling the interactions between an agent and its environment makes it possible to understand how the agent's decisions are made and what impact they have on the environment. By incorporating these interactions into an explanatory model, one can better grasp the complex relationships between the agent and its environment, thereby identifying the determining factors in the decision-making process. By explicitly handling the agent-environment interaction, this type of explanation aims to explain the agent's strategy. The families of methods used to obtain explanations in the form of agent-environment interactions are: *Bayesian network learning* and *LS usage for MDP generation*.

Bayesian Network Learning

The evolution of an agent within an environment can be observed as a set of random variables connected through a Bayesian network [Madumal et al., 2020]. Random variables are defined to represent both external aspects (the environment) and internal aspects of the agent (the actions). By analyzing trajectories, it becomes possible to infer causal relationships among these random variables. At the end of this process, a weighted causal graph is obtained, which serves as an explanation for the observer, as illustrated for the video game *StarCraft II* in Figure 1.8, where nodes represent random variables and edges represent causal relationships between them. This *Bayesian network*, representing the interactions between the agent and the environment, is then used as the explanation.

Latent Space Usage for MDP Generation

Each layer of a NN can be viewed as a projection from one space to another [Zahavy et al., 2017]. As information propagates through successive layers of a NN, it becomes increasingly abstract. It can therefore be assumed that two vectors that are close in this projected space are also close in terms of the agent's perception. Based on this observation, it is possible to redefine an MDP by leveraging the abstract representations produced by the final layers of a NN. Once the new MDP has been constructed, a model of the interaction between

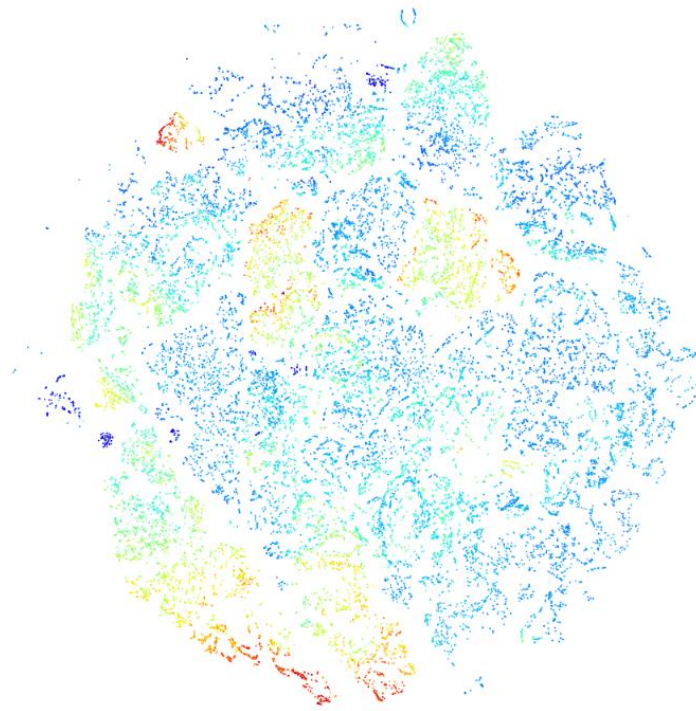


Figure 1.9: t -SNE projection of a neural network's latent space [Zahavy et al., 2017].

the agent and its environment is obtained. This model, represented as a *graph*, is then used as the explanation.

Figure 1.9 illustrates this idea using a two-dimensional t -SNE projection of the LS of the NN used to represent the policy. Each point is colored according to the value function's prediction for the corresponding state, with warmer colors indicating higher predicted values. An agreement can be observed between the structure of this projected space and the critic's predictions: states that are assigned similar values by the critic also tend to be located close to one another in the projection. This agreement makes it possible to manually identify clusters of states, which likely correspond to states that are perceived as similar by the agent.

1.4 Conclusion

The taxonomy and the state-of-the-art review presented in this chapter make it possible to draw three main observations.

1.4.1 Absence of Explanation of the Learning Process

Regarding the subject of explanation, Table 1.1 shows that none of the reviewed methods addresses the learning process itself. Such an explanation would concern

507 how the policy is progressively updated throughout training, and why it evolves
 508 over successive iterations to converge toward a particular solution. Several
 509 reasons explain this absence.

510 First, among the different machine learning paradigms, RL is arguably the
 511 one that would benefit the most from explanations of the learning process.
 512 Unlike supervised or unsupervised learning, it relies on an explicit exploration–
 513 exploitation trade-off, making the learning trajectory itself an informative subject
 514 of explanation. However, most XRL methods are inherited from the XAI
 515 literature, where explanations focus exclusively on the behavior of already-
 516 trained models. This limitation has naturally carried over to XRL.

517 Second, from a practical perspective, explanations of the learning process
 518 have limited interest. In most real-world applications, practitioners deploy an
 519 already-trained agent, while training itself takes place offline. Users are generally
 520 interested in understanding the behavior of the final policy rather than the
 521 optimization process that produced it.

522 Finally, explaining the learning process raises significant methodological
 523 challenges. Training in RL can be viewed as the iterative improvement of a policy
 524 through repeated interactions with the environment. In this sense, explaining
 525 learning would amount to explaining the evolution of an entire sequence of
 526 policies rather than a single one. Current XRL methods still struggle to provide
 527 comprehensive explanations of an individual policy, so extending them to a full
 528 training trajectory appears considerably more difficult.

529 1.4.2 Lack of Unified Metrics for Explainability

530 The diversity of explanation forms identified in this review also reveals another
 531 limitation shared by nearly all families of XRL methods, which is the absence
 532 of reliable metrics for evaluating explainability. In many studies, explanations
 533 are not evaluated with human participants. Instead, authors focus on technical
 534 properties of the proposed method, such as fidelity, sparsity, or computational
 535 efficiency. When human evaluations are conducted, they usually rely on subjective
 536 satisfaction questionnaires. Although useful, self-reported satisfaction remains
 537 prone to numerous biases and does not necessarily indicate that the explanation
 538 has improved the observer’s mental model or understanding of the agent.

539 This limitation is not specific to XRL, it reflects a broader challenge affecting
 540 XAI as a whole. In our view, this limitation comes from how explainability has
 541 developed in artificial intelligence. Most of the proposed approaches have been
 542 developed by the computer science community. Explainability has consequently
 543 often been studied from an algorithmic or mathematical perspective, with less
 544 attention given to how humans actually benefit from explanations.

545 Our definition of explanation explicitly places the observer at the center of the
 546 explanatory process. From this perspective, the value of an explanation should
 547 not only depend on its technical properties but also on its ability to improve
 548 the observer’s understanding. Addressing this challenge will require stronger
 549 interdisciplinary collaboration between computer science and disciplines such
 550 as cognitive science, psychology, philosophy, and sociology. These fields provide
 551 the theoretical and experimental tools needed to study human understanding
 552 and to develop evaluation protocols that measure the actual effectiveness of
 553 explanations.

554 1.4.3 Toward Intrinsic Explainability in XRL

555 Our review also suggests that current XRL research remains largely dominated
 556 by substitutive explainability approaches. Replacing a deep model with a
 557 simpler one therefore risks losing precisely the complexity that makes deep RL
 558 effective. For this reason, we argue that a promising direction for XRL is to
 559 develop explanations directly from the NN that forms the agent’s policy. Such
 560 approaches aim to analyze the representations learned by the network itself.
 561 From this perspective, the NN should not necessarily be regarded as an black
 562 box, but as a complex representation whose apparent opacity may instead reflect
 563 the limitations of current analysis tools.

564 Among the methods reviewed, *latent space usage for MDP generation* (Sec-
 565 tion 1.3.6) appears closest to this intrinsic perspective. However, this approach
 566 still relies on a largely manual and qualitative analysis of the LS, which limits
 567 its reproducibility and its ability to produce systematic explanations. The next
 568 chapter builds upon this idea and proposes a mechanistic XRL approach based
 569 on LS analysis.

570 Acronyms

571 **AI** Artificial Intelligence 1, 5, 6

572 **DL** Deep Learning 1, 5, 6

573 **LLM** Large Language Model 1

574 **LR** Latent Representation 13

575 **LS** Latent Space 13, 16, 17, 19

576 **MDP** Markov Decision Processes 10, 16

577 **NN** Neural Network 5, 9, 10, 12, 16, 17, 19

578 **RL** Reinforcement Learning 1, 2, 6–11, 14, 18, 19

579 **XAI** eXplainable Artificial Intelligence 1, 2, 5–7, 18

580 **XRL** eXplainable Reinforcement Learning 1, 2, 6–10, 14, 18, 19

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